



## **UE23CS352B - Object Oriented Analysis & Design**

### **Mini Project Report**

### **Event Budget Planning and Expense Approval System**

*Submitted by:*

*Team Members*

| NAME                 | SRN           |
|----------------------|---------------|
| C Kaustubh           | PES1UG23CS154 |
| Chiyedu Vishnu       | PES1UG23CS169 |
| Dareddy Devesh Reddy | PES1UG23CS171 |
| BB Varun Kumar       | PES1UG23CS131 |

Semester - Section

**Faculty Name - Shridevi Sawant**

**January - May 2026**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**FACULTY OF ENGINEERING**

**PES UNIVERSITY**

(Established under Karnataka Act  
No. 16 of 2013) 100ft Ring Road,  
Bengaluru – 560 085, Karnataka, India

## Problem Statement:

Event management at PES University involves multiple stakeholders — organizers who plan events, approving authorities who review budgets, and a finance office that oversees expenditure. Currently this process is entirely manual and error-prone: budget requests are submitted informally, approval routing is inconsistent, and there is no audit trail of decisions made.

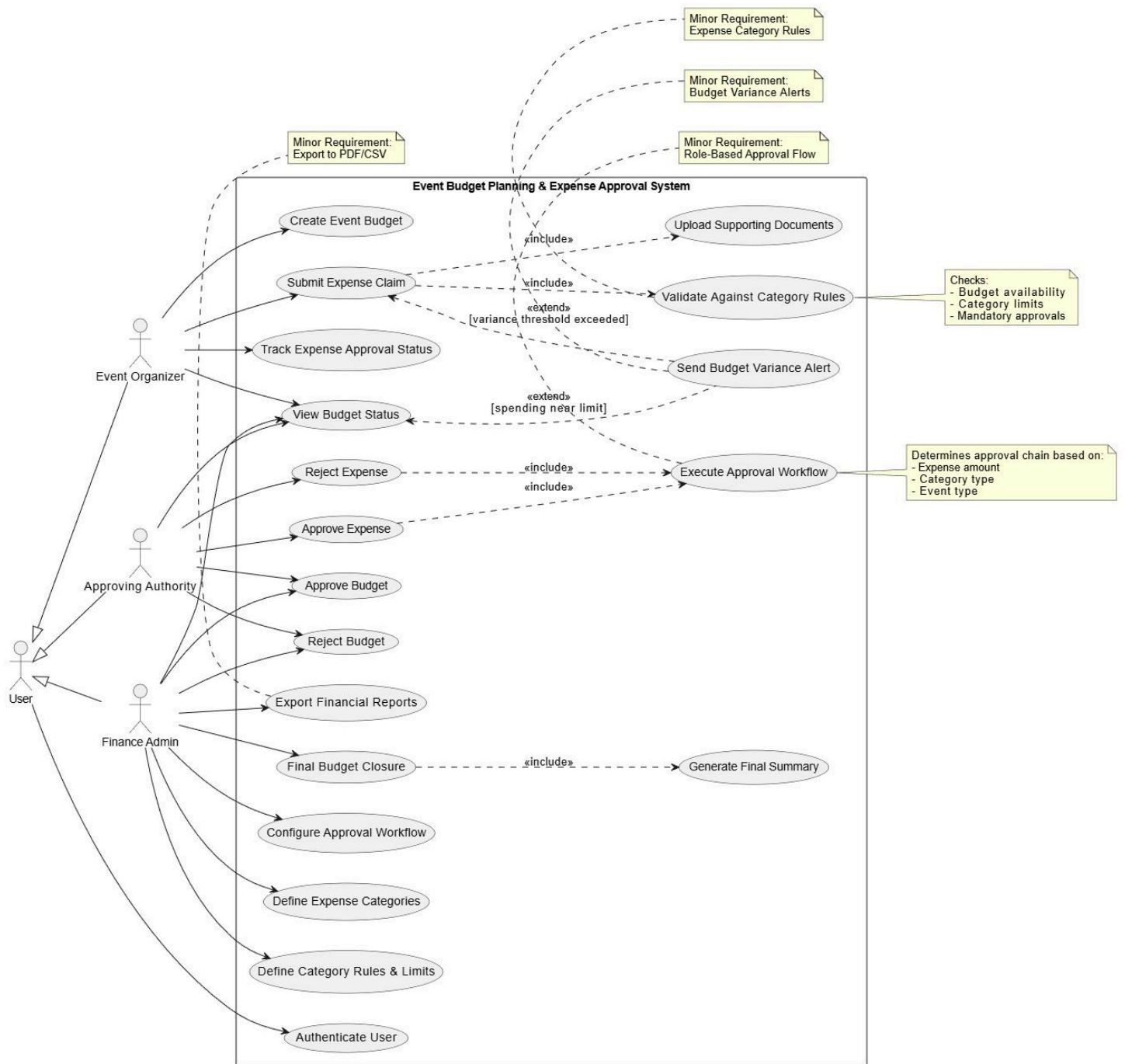
EventBudget is a web-based Event Budget Planning and Expense Approval System that digitises this workflow end-to-end. Organizers create events and submit structured budgets broken down by expense category. Approval is automatically routed based on configurable thresholds (auto-approve, single-level, or multi-level). Expense claims are tracked against approved budgets with real-time balance monitoring. The Finance Admin can close budgets and export financial reports in CSV or PDF format. The system ensures accountability, enforces spending limits, and provides a complete audit trail of every decision.

## Key Features:

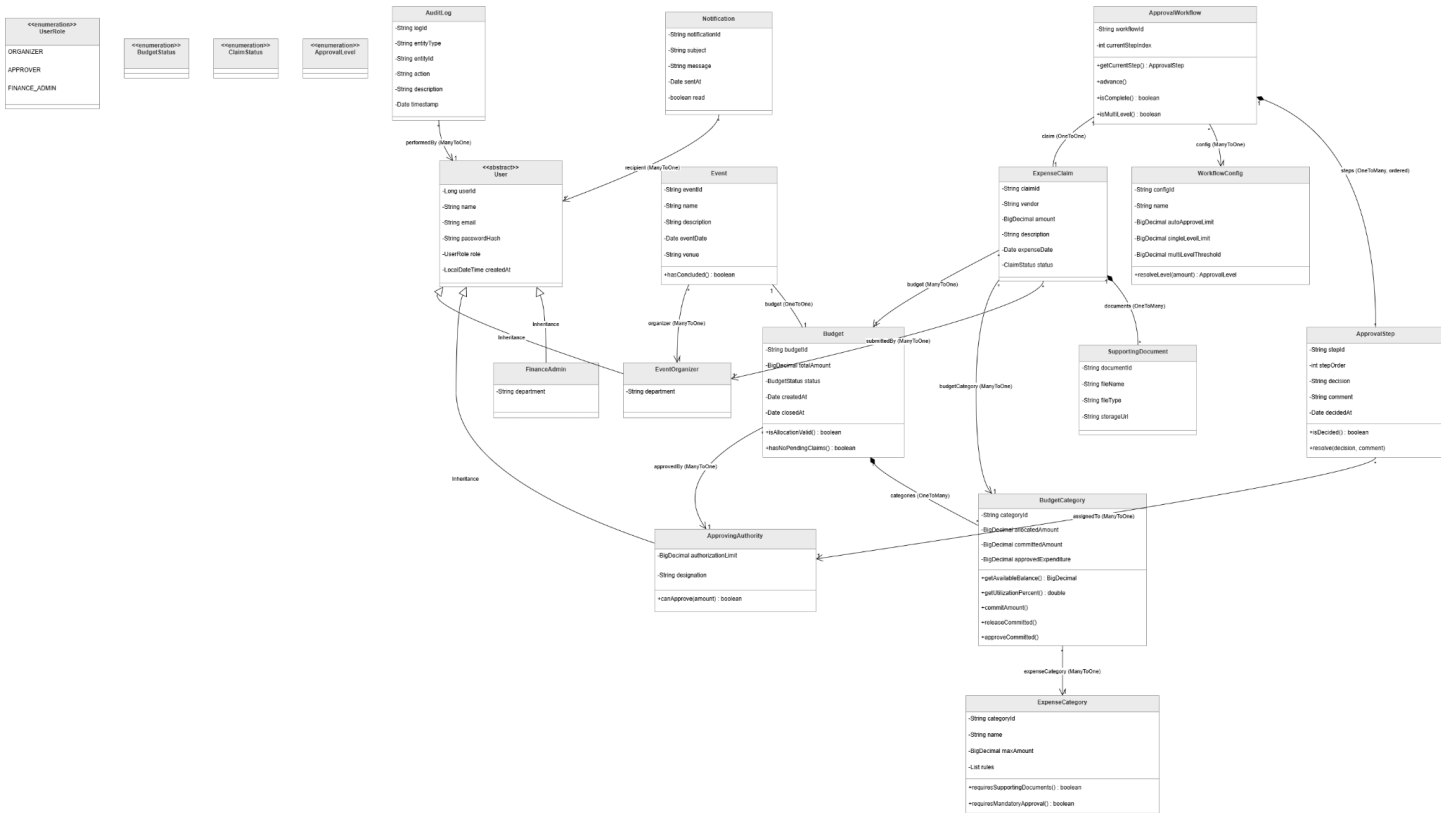
- **Role-Based Access Control** — Three distinct roles: Event Organizer, Approving Authority (Faculty Coordinator / Dean), and Finance Admin, each with a dedicated dashboard and restricted API endpoints secured via JWT.
- **Event Management** — Organizers create and manage events with name, date, venue, and description. Each event can have exactly one budget.
- **Budget Planning with Category Allocation** — Budgets are decomposed into expense categories (Venue, Food, Marketing, Travel). The sum of category allocations must exactly equal the declared total amount before submission.
- **Configurable Approval Workflow** — Finance Admin sets three thresholds: Auto-Approve Limit, Single-Level Limit, and Multi-Level Threshold. Claims are routed automatically based on the claim amount against the active WorkflowConfig.
- **Multi-Tier Expense Approval** — Auto-approved ( $\text{₹} \leq 0$ ), single-level ( $\text{₹} 0 - \text{₹} 10,000$ ), or multi-level ( $> \text{₹} 10,000$ ) approval chains depending on WorkflowConfig and per-category rules (e.g., Travel always requires mandatory approval).
- **Expense Claim Submission** — Organizers submit claims against approved budget categories with vendor, amount, expense date, and optional supporting documents. Travel and Venue categories require mandatory supporting documentation.
- **Real-Time Budget Utilisation Tracking** — Each BudgetCategory tracks committed (pending) and approved expenditure separately. Available balance is computed live and prevents over-spending.
- **Notification System** — In-app notifications are dispatched to relevant users at every workflow transition: budget submitted, approved, or rejected; claim submitted, auto-approved, escalated, approved, or rejected.
- **Audit Trail** — Every significant action (budget approval, claim decision, budget closure, workflow config creation) is recorded with actor, timestamp, entity reference, and description.
- **Financial Reporting** — Finance Admin can export any budget as a structured CSV (full category breakdown with utilisation %) or a plain-text PDF report.

## Models:

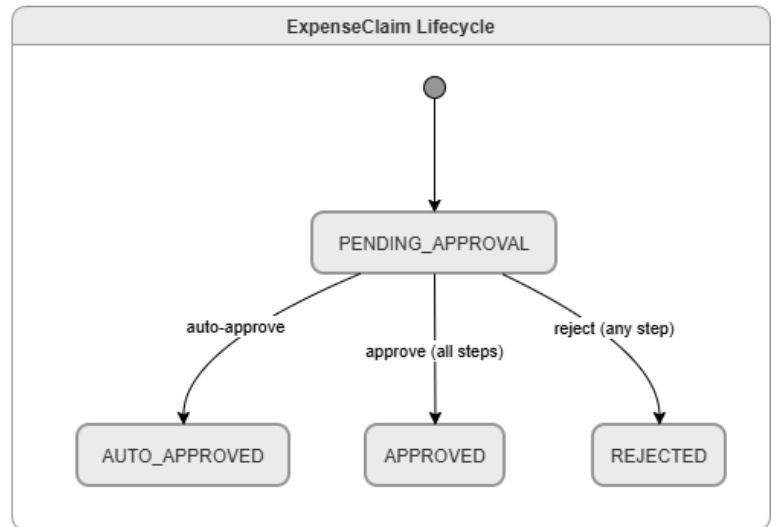
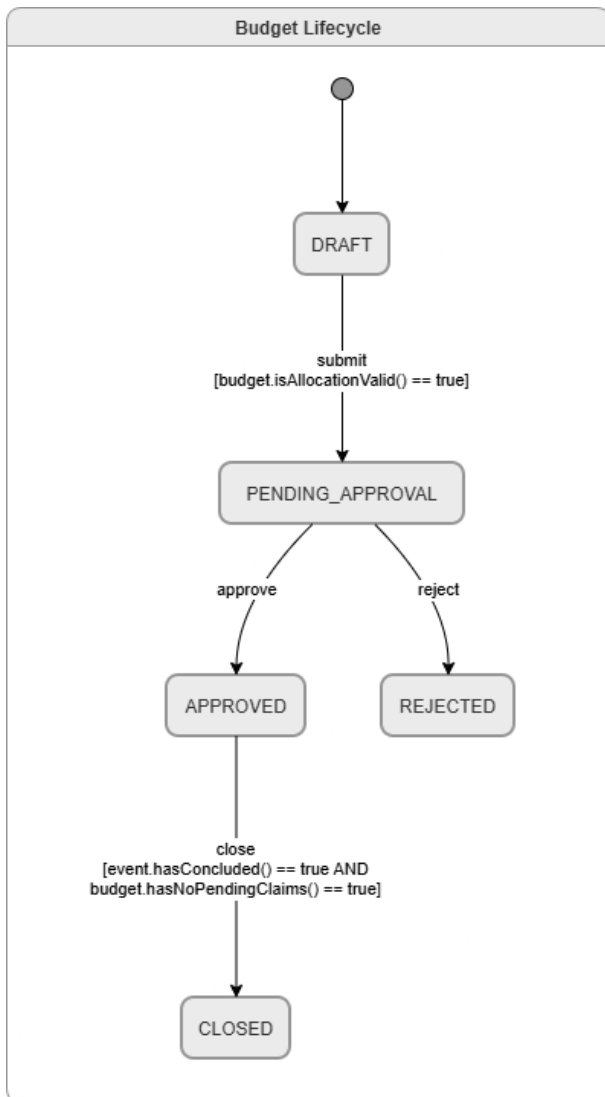
### Use Case Diagram:



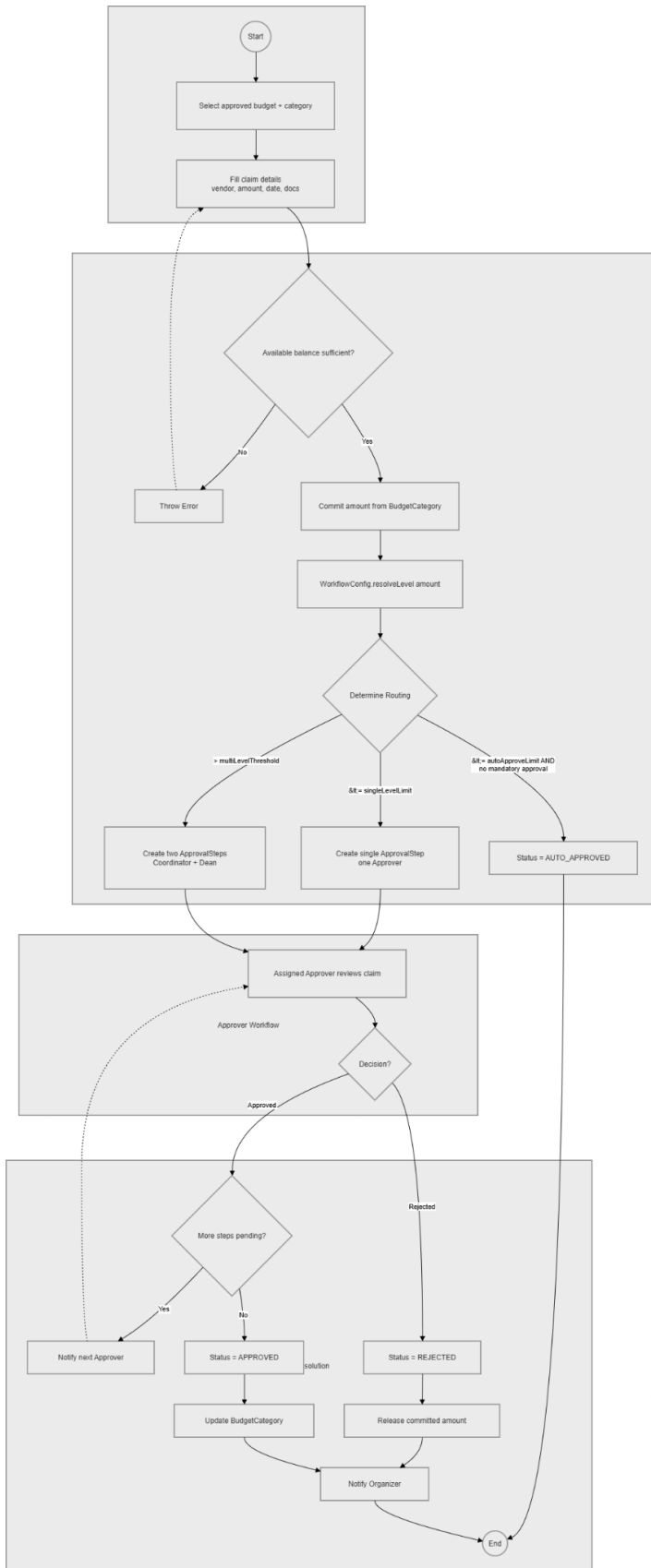
# Class Diagram:

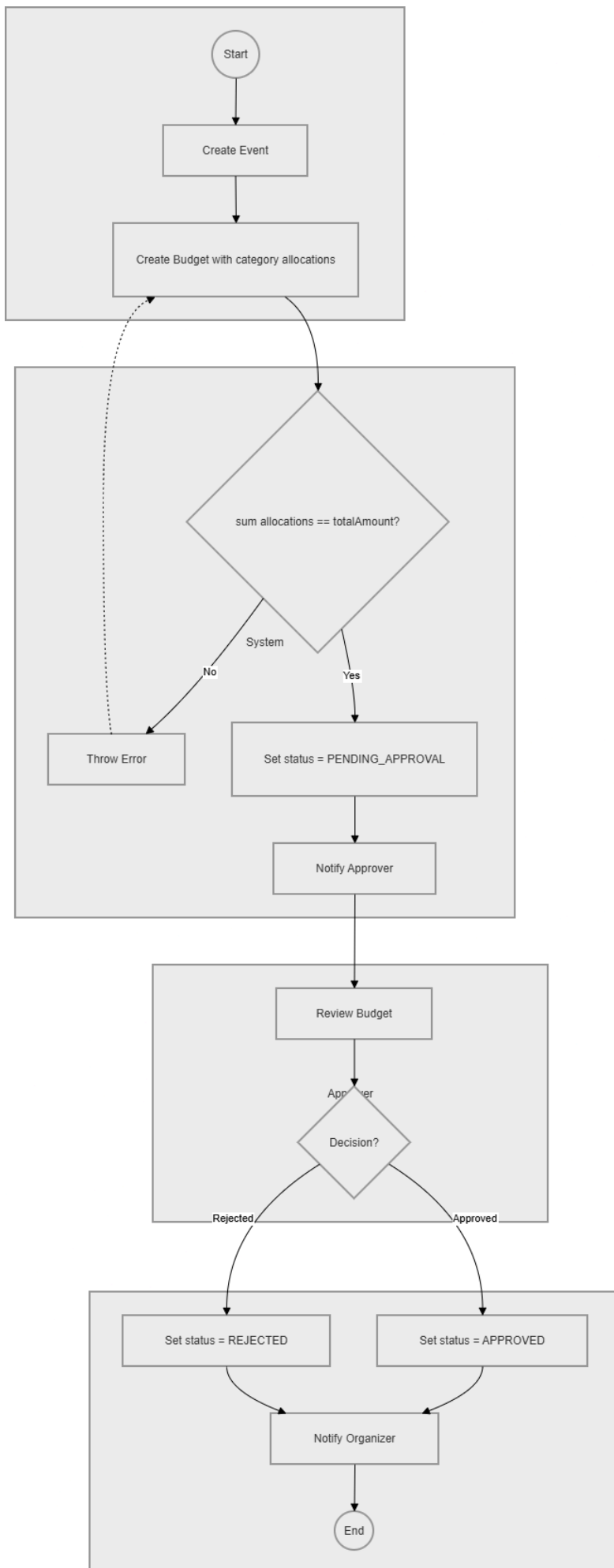


## State Diagram:



## Activity Diagrams:





## Design Principles, and Design Patterns:

### MVC Architecture used? Yes/No

YES

| Layer                                  | Classes                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model</b>                           | All classes under the <code>model/</code> package (User hierarchy, Budget, ExpenseClaim, etc.)                                         |
| <b>Controller</b><br>(View-Controller) | <code>AuthController</code> , <code>OrganizerController</code> , <code>ApproverController</code> , <code>FinanceController</code>      |
| <b>Service</b>                         | <code>OrganizerService</code> , <code>ApproverService</code> , <code>FinanceService</code> , <code>AuthService</code> (business logic) |
| <b>Repository</b>                      | Spring Data JPA repositories (data access layer)                                                                                       |

**Note:** The frontend SPA (`app.js`) acts as a thin **View**, consuming the REST API.

### Design Principles

#### Single Responsibility Principle (SRP)

The Single Responsibility Principle (SRP) dictates that each service class has exactly one responsibility. In this architecture, the `OrganizerService` handles only organizer operations, the `NotificationService` handles only notification delivery, and the `AuditService` handles only audit logging. Furthermore, the controllers strictly handle HTTP concerns and delegate all business logic to their respective services.

#### Open/Closed Principle (OCP)

The Open/Closed Principle (OCP) ensures that the approval routing system is open for extension but closed for modification. For example, adding a new approval tier like a `COMMITTEE_LEVEL` requires only implementing a new `ApprovalRoutingStrategy` and registering it, which results in absolutely no changes to existing code. The `ApprovalWorkflowFactory` then automatically discovers these strategies via Spring's



dependency injection.

### **Liskov Substitution Principle (LSP)**

The Liskov Substitution Principle (LSP) is demonstrated through the inheritance hierarchy formed by User, EventOrganizer, ApprovingAuthority, and FinanceAdmin. Any method in the system that accepts a base User object will work correctly when given any of these subtypes. Specifically, the ApprovingAuthority's canApprove method extends the base User functionality without breaking its underlying contract.

### **Dependency Inversion Principle (DIP)**

The Dependency Inversion Principle (DIP) is applied by ensuring all services depend on repository interfaces, such as Spring Data JPA, rather than concrete implementations. Similarly, the ApprovalWorkflowFactory depends entirely on the ApprovalRoutingStrategy interface rather than directly relying on any concrete strategy classes.

## **Design Patterns**

### 1. Strategy Pattern

- Intent: Define a family of algorithms, encapsulate each one, and make them interchangeable.
- Where used: Approval routing in com.eventbudget.service.strategy
- Interface: ApprovalRoutingStrategy — defines supports(): ApprovalLevel and createWorkflow(...): ApprovalWorkflow
- Concrete Strategies:
  - AutoApprovalStrategy — instantly approves the claim, no steps created
  - SingleLevelApprovalStrategy — creates one ApprovalStep assigned to the most capable approver
  - MultiLevelApprovalStrategy — creates two sequential ApprovalSteps (Coordinator, then Dean)
- Context: ApprovalWorkflowFactory selects the strategy based on WorkflowConfig.resolveLevel(amount)
- Benefit: Adding a new approval tier requires no changes to existing strategies or the factory dispatch logic.

### 2. Factory Pattern

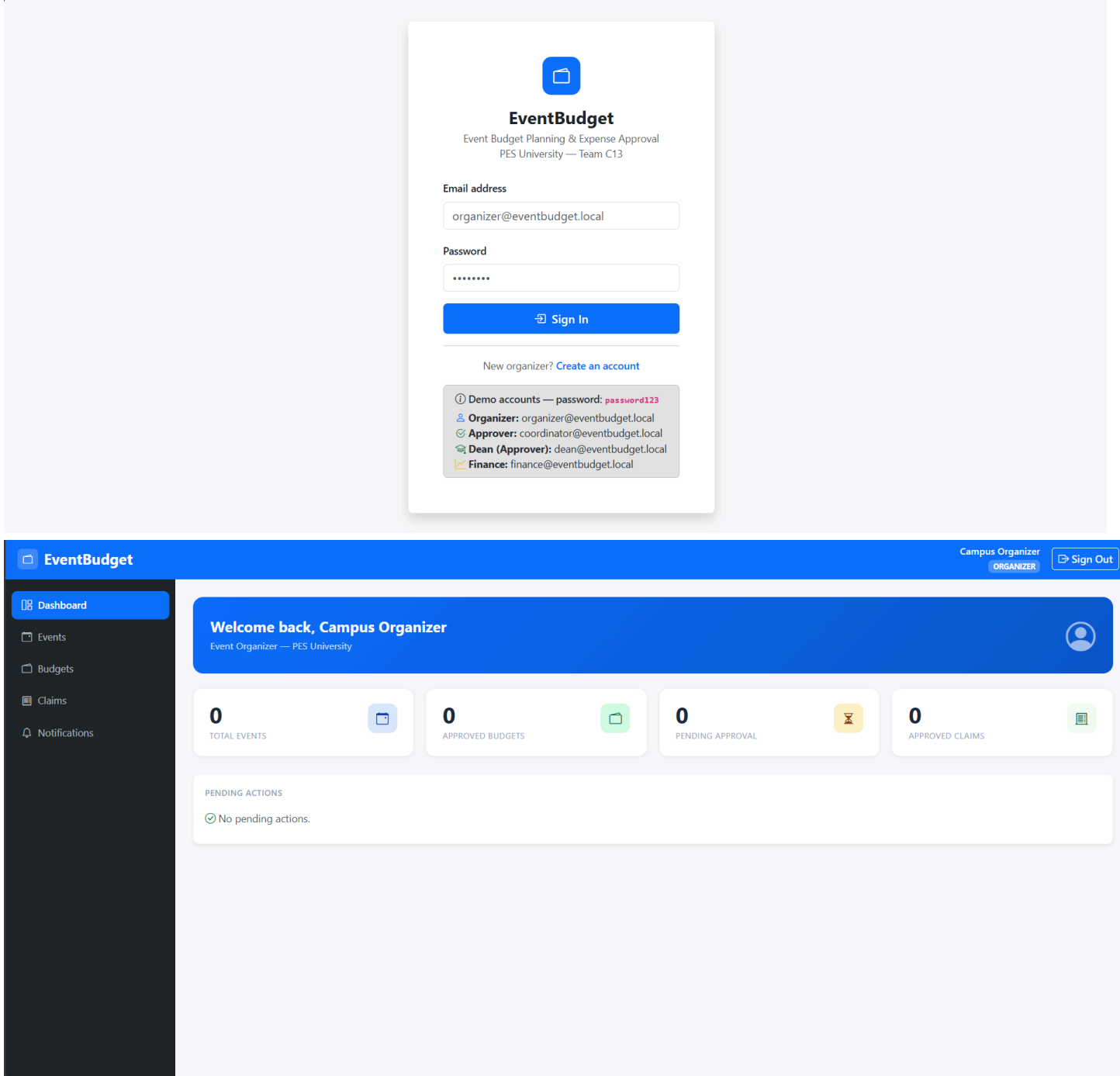
- Intent: Encapsulate object creation logic behind a single interface.
- Where used: ApprovalWorkflowFactory in com.eventbudget.service.factory
- How it works: The factory receives all ApprovalRoutingStrategy beans from Spring, stores them in an EnumMap<ApprovalLevel, ApprovalRoutingStrategy>, and exposes a single createWorkflow(claim, config, approvers) method. Callers do not need to know which strategy is used — the factory resolves the correct one from the amount.
- Benefit: Decouples the OrganizerService (which triggers claim creation) from the approval routing logic entirely.

**Github link to the Codebase: (repository should be public)**

<https://github.com/snigenigmatic/eventbudget>

# Screenshots

UI:



EventBudget

Campus Organizer  
ORGANIZER

Sign Out

Dashboard

Events

Budgets

Claims

Notifications

My Events

+ New Event

New Event

Event Name

Tech Symposium 2025

Description

Brief description

Event Date

dd-mm-yyyy

Venue

Seminar Hall A

Cancel

Create Event

EventBudget

Campus Organizer  
ORGANIZER

Sign Out

Dashboard

Events

Budgets

Claims

Notifications

My Events

+ New Event

| # | NAME                | DESCRIPTION   | DATE      | VENUE          | BUDGET    |
|---|---------------------|---------------|-----------|----------------|-----------|
| 1 | Tech Sumposiym 2025 | it is a event | 17/4/2026 | Seminar Hall A | No budget |

Event created!

EventBudget

Campus Organizer  
ORGANIZER

Sign Out

Dashboard

Events

Budgets

Claims

Notifications

My Budgets

+ New Budget

New Budget

Event  
Tech Sumposiym 2025 (17/4/2026)

Category Allocations — total will be computed from these

Venue (max: ₹20,000.00)

10000

Food (max: ₹15,000.00)

10000

Marketing (max: ₹8,000.00)

5000

Travel (max: ₹25,000.00)

10000

+ Add Category

Total: ₹35,000.00

Cancel

Submit for Approval

EventBudget

Campus Organizer  
ORGANIZER

Sign Out

Dashboard

Events

Budgets

Claims

Notifications

My Budgets

+ New Budget

| # | EVENT               | TOTAL      | ALLOCATED  | STATUS           | APPROVED BY | CREATED                |
|---|---------------------|------------|------------|------------------|-------------|------------------------|
| 1 | Tech Sumposiym 2025 | ₹35,000.00 | ₹35,000.00 | PENDING APPROVAL | —           | 15/4/2026, 11:20:37 pm |

EventBudget

Faculty CoordinatorAPPROVERSign Out

Dashboard

Pending Budgets

Pending Claims

Notifications

Welcome back, Faculty Coordinator

Approving Authority — PES University

1

BUDGETS TO REVIEW

0

CLAIMS TO REVIEW

1 item(s)

require your review. Check the tabs above.

EventBudget

Faculty CoordinatorAPPROVERSign Out

Dashboard

Pending Budgets

Pending Claims

Notifications

Pending Budgets

| # | EVENT               | TOTAL      | CATEGORIES                                                                          | SUBMITTED              | ACTION                                   |
|---|---------------------|------------|-------------------------------------------------------------------------------------|------------------------|------------------------------------------|
| 1 | Tech Sumposiym 2025 | ₹35,000.00 | Venue: ₹10,000.00<br>Food: ₹10,000.00<br>Marketing: ₹5,000.00<br>Travel: ₹10,000.00 | 15/4/2026, 11:20:37 pm | <div>✓ Approve</div> <div>✗ Reject</div> |

- Dashboard
- Events
- Budgets
- Claims
- Notifications

My Claims

+ Submit Claim

| # | CATEGORY | VENDOR           | AMOUNT    | DATE      | STATUS   | APPROVAL LEVEL |
|---|----------|------------------|-----------|-----------|----------|----------------|
| 1 | Food     | City Travels Ltd | ₹1,500.00 | 15/4/2026 | APPROVED | AUTO APPROVED  |

- Dashboard
- Workflow Config
- All Budgets
- Audit Log
- Notifications

Welcome back, Finance Office

Finance Administrator — PES University

1

TOTAL BUDGETS

1

APPROVED & OPEN

0

CLOSED

4

AUDIT EVENTS

EventBudget

Finance Office  
FINANCE ADMIN

Sign Out

Dashboard

Workflow Config

All Budgets

Audit Log

Notifications

Workflow Configurations

+ New Config

Thresholds must satisfy: **Auto-Approve** ≤ **Single-Level** ≤ **Multi-Level**

| # | NAME                    | AUTO-APPROVE LIMIT | SINGLE-LEVEL LIMIT | MULTI-LEVEL THRESHOLD |
|---|-------------------------|--------------------|--------------------|-----------------------|
| 1 | Default Approval Policy | ₹2,000.00          | ₹10,000.00         | ₹10,000.00            |

EventBudget

Finance Office  
FINANCE ADMIN

Sign Out

Dashboard

Workflow Config

All Budgets

Audit Log

Notifications

Workflow Configurations

+ New Config

Thresholds must satisfy: **Auto-Approve** ≤ **Single-Level** ≤ **Multi-Level**

| # | NAME                    | AUTO-APPROVE LIMIT | SINGLE-LEVEL LIMIT | MULTI-LEVEL THRESHOLD |
|---|-------------------------|--------------------|--------------------|-----------------------|
| 1 | Default Approval Policy |                    |                    | ₹10,000.00            |

New Workflow Config

Configuration Name

Standard 2025

Auto-Approve Limit (₹)

500

Single-Level Limit (₹)

5000

Multi-Level Threshold (₹)

20000

Cancel

Save Config

EventBudget

Finance Office  
FINANCE ADMIN

Sign Out

Dashboard

Workflow Config

All Budgets

Audit Log

Notifications

Audit Log

| # | ENTITY | ID | ACTION    | DESCRIPTION                                    | PERFORMED BY        | TIMESTAMP              |
|---|--------|----|-----------|------------------------------------------------|---------------------|------------------------|
| 4 | CLAIM  | 1  | SUBMITTED | Submitted expense claim for 1500               | Campus Organizer    | 15/4/2026, 11:25:11 pm |
| 3 | BUDGET | 1  | APPROVED  | —                                              | Faculty Coordinator | 15/4/2026, 11:21:57 pm |
| 2 | BUDGET | 1  | CREATED   | Submitted budget for event Tech Sumposiym 2025 | Campus Organizer    | 15/4/2026, 11:20:37 pm |
| 1 | EVENT  | 1  | CREATED   | Created event Tech Sumposiym 2025              | Campus Organizer    | 15/4/2026, 11:18:50 pm |

EventBudget

Finance Office  
FINANCE ADMIN

Sign Out

Dashboard

Workflow Config

All Budgets

Audit Log

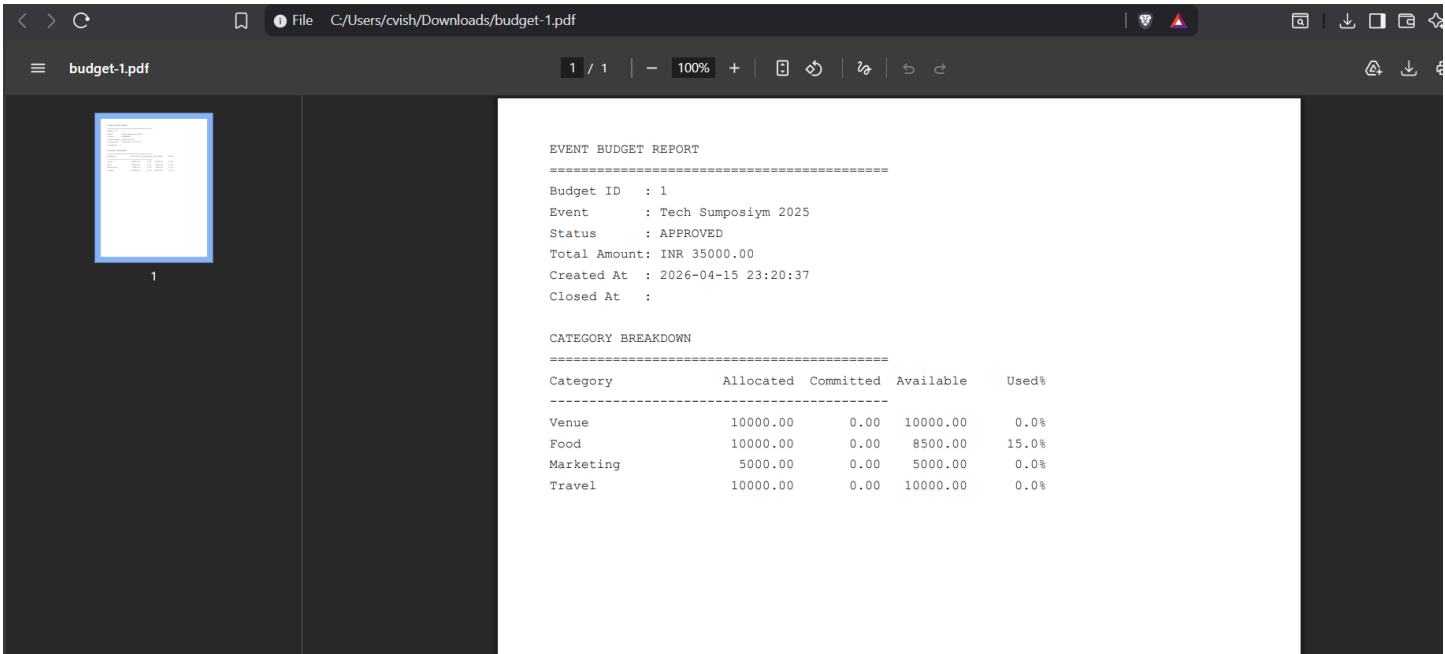
Notifications

All Budgets

| # | EVENT               | TOTAL      | STATUS   | APPROVED BY         | ACTIONS                                        |
|---|---------------------|------------|----------|---------------------|------------------------------------------------|
| 1 | Tech Sumposiym 2025 | ₹35,000.00 | APPROVED | Faculty Coordinator | <div>Close</div> <div>CSV</div> <div>PDF</div> |

Export downloaded!





Clipboard

Font

Alignment

A1

✕ ✓ *fx*

Budget ID

|    | A         | B         | C        | D         | E          | F             | G | H |
|----|-----------|-----------|----------|-----------|------------|---------------|---|---|
| 1  | Budget ID | Event     | Status   | Total Amo | Created At | Closed At     |   |   |
| 2  | 1         | Tech Sump | APPROVED | 35000     | #####      |               |   |   |
| 3  |           |           |          |           |            |               |   |   |
| 4  | Category  | Allocated | Committe | Approved  | Available  | Utilization % |   |   |
| 5  | Venue     | 10000     | 0        | 0         | 10000      | 0             |   |   |
| 6  | Food      | 10000     | 0        | 1500      | 8500       | 15            |   |   |
| 7  | Marketing | 5000      | 0        | 0         | 5000       | 0             |   |   |
| 8  | Travel    | 10000     | 0        | 0         | 10000      | 0             |   |   |
| 9  |           |           |          |           |            |               |   |   |
| 10 |           |           |          |           |            |               |   |   |

Individual contributions of the team members:

| <b>Name</b>                 | <b>Module worked on</b>      |
|-----------------------------|------------------------------|
| <b>C Kaustubh</b>           | Auth, Security & Users       |
| <b>Chiyedu Vishnu</b>       | Events & Budgets (Organizer) |
| <b>Dareddy Devesh Reddy</b> | Approval Workflow            |
| <b>BB Varun Kumar</b>       | Finance, Claims & Reports    |